

PROBLEM STATEMENT

AI-Based Predictive Modelling for Early Forecasting of Bovine Mastitis in Indian Dairy Farms

1. Industry Context

The Indian dairy sector plays an important role in food production, rural livelihoods and agricultural income. Dairy farms range from small household-level units to organized commercial farms and differ in herd size, animal breeds, management practices, hygiene conditions and access to veterinary services. Maintaining animal health is therefore essential for stable milk production and farm profitability.

Bovine mastitis is a major health and production concern in dairy cattle. It is an inflammatory condition of the mammary gland and can be associated with infectious and management-related factors. Mastitis can reduce milk production, affect milk quality, increase veterinary and treatment costs, and contribute to economic losses. Clinical cases may show visible signs, while subclinical cases can be difficult to recognize because obvious symptoms may be absent.

The risk of mastitis can be influenced by previous mastitis history, lactation stage, parity, milk yield, udder condition, milking hygiene, housing conditions, environmental temperature and humidity, cleanliness of animals and equipment, and other management factors. These factors can change over time, so monitoring only one factor may not provide sufficient information for early forecasting.

Current monitoring practices can depend heavily on farmer observation, routine records, milk-quality checks and veterinary examination after a problem becomes noticeable. These approaches remain important, but they can become difficult to manage when many animals are present or records are maintained manually. Historical information may contain useful patterns that are not easily recognized through manual review.

2. Problem Statement

Design and develop an AI-Based Predictive Modelling system for early forecasting of bovine mastitis risk in Indian dairy farms. The system should collect and process relevant animal, milk, environmental and farm-management data and use machine-learning techniques to estimate the likelihood of mastitis for individual animals or farm records.

The platform should support the complete predictive workflow, including data collection, validation, preprocessing, feature selection, model training, model evaluation, risk prediction and visualization. Historical records containing known mastitis outcomes should be used to train and evaluate suitable predictive models.

The system should identify patterns associated with increased mastitis risk before the condition becomes clearly observable wherever the available data supports such forecasting. The output may be represented as a probability or understandable risk categories such as low, medium and high.

The platform should provide an easy-to-understand interface through which authorized users can enter or review animal information, view predicted risk, examine historical records and identify animals requiring closer monitoring. Higher-risk cases may generate alerts.

The system should record follow-up outcomes so predicted results can be compared with later observations or confirmed cases. This information can support model evaluation and future improvement. Data quality, privacy, security and responsible use should be considered throughout the system.

The solution is not intended to replace veterinary professionals or laboratory diagnosis. It provides an early-warning mechanism that helps prioritize attention and supports data-informed farm management.

3. Objectives

- Collect relevant animal, milk, environmental and farm-management data.
- Create and maintain a structured dataset for mastitis prediction.
- Preprocess data by handling missing, duplicate and inconsistent records.
- Identify important features associated with mastitis risk.
- Develop suitable machine-learning models for mastitis prediction.
- Evaluate and compare model performance using appropriate metrics.
- Predict mastitis risk for individual animals.
- Provide a simple dashboard for displaying prediction results.
- Maintain historical prediction and animal-health records.
- Support early monitoring and veterinary decision-making through risk alerts.

4. Expected Outcomes:

- Develop an end-to-end AI-Based Bovine Mastitis Prediction System for collecting, processing and analyzing dairy-farm data.
- Provide a GUI/dashboard for entering animal information, viewing predictions and visualizing mastitis-risk results.
- Maintain information such as animal ID, age, breed, parity, lactation stage, milk yield, previous mastitis history, environmental conditions and health records.
- Enable analysis and correlation of animal, milk, environmental and management factors for better mastitis-risk assessment.
- Provide secure and efficient data storage, retrieval and management of animal and prediction records.
- Generate individual animal mastitis-risk predictions and risk categories to support early monitoring.
- Apply software engineering principles such as security, scalability, maintainability, reliability and usability.
- Demonstrate proper requirements analysis, system design, AI model development, evaluation, limitations and future enhancements.